YAGO

YAGO: What is it? 1
How YAGO differs from DBpedia 1
Annotation via YAGO 1
Annotating with YAGO in the NLP Suite 2
Further readings — more TIPS files 2

YAGO: What is it?

YAGO ("Yet Another Great Ontology") is a large, open knowledge base developed at the Max Planck Institute for Informatics and Télécom Paris. Like DBpedia, it extracts structured, machine-readable knowledge from Wikipedia — but it combines Wikipedia's breadth with the clean, hand-checked taxonomy of WordNet and the place data of GeoNames, and from YAGO 4 onward draws its facts from Wikidata. The result is a knowledge graph in which millions of entities — people, places, organizations, works, events — are typed according to a consistent class hierarchy.

How YAGO differs from DBpedia

- Both **DBpedia** and **YAGO** extract structured knowledge from Wikipedia / Wikidata and can annotate the entities in your corpus.
- **DBpedia** organizes entities with its own DBpedia ontology. It lets you set a confidence level (via DBpedia Spotlight).
- **YAGO** types its entities using classes from schema.org, the same vocabulary search engines use to describe web content. That is why YAGO's ontology classes in the NLP Suite are the schema.org classes. **YAGO** does not use the complete schema.org class set — the NLP Suite works with a reduced list. For the classes actually available, see TIPS_NLP_Annotator YAGO (schema reduced).pdf; for the full schema.org listing, see TIPS_NLP_Annotator YAGO (schema.org) ontology classes.pdf.

Annotation via YAGO

To annotate a text is to attach machine-readable information to the words in it while leaving the text itself untouched — a layer of description laid over the running prose, saying of a given string what kind of thing it refers to.

Named entity recognition already does part of this: it tells you that Atlanta is a place and Jim Cobb a person. Annotating against a knowledge graph goes one step further. Rather than assigning a type from a short list of tags, it LINKS the string in your document to a particular entity in the knowledge base, and inherits that entity's class in the ontology. Atlanta is then tied to the city in Georgia rather than to the film, the ship, or anything else bearing the name, and with the link come the facts the knowledge base holds about it — its coordinates, the country it belongs to, the classes it falls under.

The payoff for research is that a corpus becomes searchable by CLASS rather than by word. You can ask for every Organization named across a thousand newspaper articles without knowing in advance which newspapers, companies or associations appear in

them, or count how often a Person is mentioned alongside a Place. Where a keyword search finds only the strings you thought to type, annotation lets you ask questions of a kind you could not otherwise pose.

Two cautions are worth keeping in mind. Annotation is a judgment made by software and it can be wrong: names are ambiguous, different people and places share them, and an entity that is missing from the knowledge base cannot be linked at all. Historical corpora suffer most here, since nineteenth-century local place names and ordinary people are thinly covered in Wikipedia and Wikidata. Treat the annotated output as a first pass to be inspected, not as a finished classification.

Annotating with YAGO in the NLP Suite

YAGO is annotated from the same GUI as DBpedia (the Knowledge graphs annotator). Select YAGO from the knowledge-base dropdown, then choose one or more ontology classes (and, if you wish, more specific sub-classes). If no class is selected, all classes are processed, with Thing as the default.

- INPUT: one or more .txt files.
- OUTPUT: annotated .html files, plus a .csv of the words matched to each class.

Further readings — more TIPS files

TIPS_NLP_Annotator DBpedia.pdf · TIPS_NLP_Annotator YAGO (schema.org) ontology classes.pdf · TIPS_NLP_Annotator YAGO (schema reduced).pdf · TIPS_NLP_W3C OWL RDF SPARQL.pdf

Links: https://yago-knowledge.org/ · http://yago.r2.enst.fr/